

Azhad Shahzad Shaik

azhadshahzads@gmail.com | +1 959-759-3928 | Hartford, CT, USA
github.com/s-shahzad | azhadshahzadshaik.netlify.app

RESEARCH INTERESTS

- **Critical-infrastructure & EV/IoT security** — protocol-level attack-surface modeling (OCPP, MQTT, CAN), ML-based DoS detection, OT/IT convergence
- **Machine learning for intrusion detection** — hybrid signature-plus-anomaly fusion, supervised (RandomForest) and unsupervised (IsolationForest) ensembles, false-positive reduction at production scale
- **Medical-device and health-IT security** — HL7/FHIR pipeline integrity, IoT-based patient monitoring, PHI-aware data engineering
- **Detection engineering at scale** — soak-tested IDS pipelines, telemetry correlation, AI-assisted alert triage and explanation
- **Supply-chain, compliance, and risk automation** — NIST CSF mapping, secure-SaaS backend engineering, governance-by-default platforms

PUBLICATIONS

1. **A. S. Shaik**, M. Gutla, A. P. R. Kagitala, and S. Bhatia, “Plugged-in and Protected: Leveraging Machine Learning to Secure IoT-Based Electric Vehicle Charging Stations from Denial-of-Service Threats,” *IEEE Global Conference on Artificial Intelligence and Internet of Things (GCAIoT)*, CT, USA, 2025. **First author**. DOI: 10.1109/GCAIoT68269.2025.11275540
 - Trained and benchmarked four supervised ML classifiers (Decision Tree, Random Forest, SVM-RBF, XGBoost) on a 1.62 M-sample CICIDS-derived EVCS network traffic dataset (benign + DoS, SMOTE-balanced, 80/20 split) to detect Denial-of-Service attacks on OCPP-protocol IoT EV charging infrastructure.
 - Random Forest best performance: 99.03% accuracy, 91.21% precision, 86.56% recall, 88.83% F1, 99.81% AUROC. Framed within critical-infrastructure protection (smart grid + clean energy) context.
2. N. P. Singh, A. Kanakamalla, **A. S. Shaik**, G. D. Sai, and S. Suman, “Remote Monitoring System of Heart Conditions for Elderly Persons with ECG Machine Using IoT Platform,” *Journal of Information Systems and Telecommunication (JIST)*, vol. 10, no. 37, pp. 11–19, Winter 2022. Published Feb 19, 2022. **Co-author**. DOI: 10.52547/jist.15692.10.37.11
 - Designed and validated an end-to-end IoT pipeline (AD8232 ECG sensor → Arduino/ESP8266 → Wi-Fi → web dashboard) for real-time unlimited-distance remote cardiac monitoring; system auto-flags anomalous ECG signals and triggers clinician appointment booking without human intervention — outperforming Bluetooth-limited (10 m) prior systems on range, automation, and cost.

RESEARCH & PROJECT EXPERIENCE

Vector — Local-first Windows Blue-team Security Platform

Independent research project · Python, FastAPI, Ollama, Gemini, OpenAI, NVIDIA, Grok, SQLite, scapy

- Designed and shipped a FastAPI control plane (~84 production files) fusing Windows host telemetry with Universal NIDS network alerts; runs SIEM-style correlation producing scored incidents (low/medium/high) with confidence ratings, evidence highlights, timelines, and suggested next steps.
- Built a multi-provider AI router (AIProviderRouter) routing analyst tasks across Ollama (local default), Gemini, OpenAI, and Grok with automatic fallback, agent-tool bridging, health-check probing, and per-task provider/model attribution.
- Engineered FastAPI security middleware stack (LocalOnly, API key validation, rate limiting at 120 req/min) and 7 API route groups; performance-tuned telemetry collection from 3,411 ms → 2,500 ms by replacing slow Get-NetAdapter with a Win32_NetworkAdapter CIM query.
- Built threat-intel ingestion pipeline consuming the curated stamparm/ipsum feed (frequency ≥3 filter, 6-hour TTL cache, idempotent persistence to data/global_blacklist.txt).

Universal NIDS — Hybrid Intrusion-Detection Platform

Independent research project · Python, scikit-learn, scapy, FastAPI, SQLite/Postgres/JSONL, Ollama, pytest

- Built a 4-modality detection fusion engine (signature 0.4 + statistical 0.2 + supervised 0.3 + unsupervised 0.1) using IsolationForest and RandomForest classifiers; behavior-based detection for campaign, exfiltration, and lateral-movement patterns.
- Achieved research Release Candidate status: 195 pytest items, 187 passing, 79.16% coverage (above a 72% enforced floor).

- Production-validated via 6-hour soak test: 87,533 flows, **0 false alerts** at peak 107.0% CPU / 409 MB RSS / 13.335 s restart latency.
- Designed bounded adversary-emulation framework (7 safe scenarios: port scan, brute force, beaconing, exfiltration, lateral sequence, protocol anomaly, campaign chain) with safety guardrails restricting execution to offline replay or explicitly-configured lab CIDRs.

IEEE GCAIoT 2025 — ML-based DoS Detection on IoT EV Charging Stations

First-author publication · Python, scikit-learn, OCPP protocol modeling

- Modeled the protocol-level attack surface of OCPP-based IoT EV charging infrastructure under DoS threat models.
- Designed, trained, and evaluated ML classifiers (Decision Tree, Random Forest, SVM-RBF, XGBoost) on 1.62 M-sample dataset; Random Forest: 99.03% accuracy, 99.81% AUROC.
- Framed within EB-2 NIW-aligned criteria: national importance (clean-energy critical infrastructure), advanced expertise, and benefit to the United States.

Career Agent, Bounty Automator, Blockchain Auditor, Asset Monitor

Independent research projects · Python, Nuclei, Gemini Flash, headless Edge, Tor, Blockchain APIs

- Career Agent: end-to-end automation scoring JDs 1–10 against candidate profile, tailoring resumes per JD, rendering PDFs via headless Microsoft Edge, dispatching via SMTP, and logging via CareerTracker.
- Bounty Automator: AI-augmented pipeline orchestrating Nuclei scans against authorized targets, enriching findings with Gemini-Flash (CVSS 3.1 severity, exploitability, remediation), audit trail to JSONL.
- Blockchain Auditor + Dark Web Intelligence Service: investigates BTC/ETH addresses surfaced by a Tor-proxy scout cycle; flags whale wallets (>100 BTC), auto-correlates findings into security-event database.

EDUCATION

Sacred Heart University — Fairfield, CT, USA

M.S. in Cyber/Computer Forensics and Counterterrorism · Mar 2025 · GPA: 3.667 / 4.0

- Coursework: digital forensics, counterterrorism, cyber risk, network security.
- Capstone: First-author IEEE GCAIoT 2025 publication — “Plugged-in and Protected: Leveraging ML to Secure IoT-Based EV Charging Stations from DoS Threats” (DOI: 10.1109/GCAIoT68269.2025.11275540).

Koneru Lakshmaiah College of Engineering — Guntur, Andhra Pradesh, India

B.Tech in Electronics and Communications Engineering · May 2021 · CGPA: 8.21 / 10

- Coursework: signal processing, embedded systems, communication networks, control systems.

PROFESSIONAL EXPERIENCE

Solstice Solutions, Inc. — Jr. Data Analyst

Hartford, CT · Oct 2025 – Present

- Build and maintain HL7 v2 to FHIR R4 integration pipelines on Azure across a 3-stage pipeline (ingest, transform, export) covering Medicaid and Medicare (Part A, B, C) transactions; pipelines maintain a 99.9% claims approval rate.
- Process and validate EDI healthcare transactions (837 claims, 835 remittance) using SQL-driven ETL; implement validation gates at each stage enforcing structural consistency and FHIR conformance.
- Develop Power BI dashboards surfacing pipeline health, transaction volumes, and data quality metrics for clinical and operational stakeholders.

Infosysnow Inc. — Cyber Security Engineer

Remote · Aug 2025 – Oct 2025

- Operated across 3 concurrent security control streams: network hardening, anomaly triage, and remediation tracking; applied Sublime Network Oasis and OSI-layer analysis to monitor traffic and support corrective actions.
- Conducted vulnerability assessments and supported penetration testing; created and updated security protocols to reduce organizational risk exposure.

iQ4 — IT Risk Analyst Intern

Remote · May 2024 – Aug 2024

- Conducted 350-hour team-based NIST CSF analysis of the Uber 2016 data exfiltration (AWS S3 + GitHub credential compromise, 57M users); mapped control failures across all 5 CSF functions (Identify, Protect, Detect, Respond, Recover); produced 15-slide remediation briefing.
- Identified critical gaps in MFA enforcement, RBAC implementation, credential management, and real-time monitoring; tagged findings to CIA risk categories.

Sprinto Technology Pvt Ltd — Associate Software Engineer

India · Apr 2023 – Dec 2023

- Backend feature development and maintenance on compliance-focused SaaS (SOC 2 / ISO 27001 family); debugging, testing, and validating components to ensure stable, secure releases.

Vodafone Idea Ltd. — Graduate Engineering Trainee → Network Engineer (Operations)

Hyderabad, India · Nov 2021 – Apr 2023

- Led and supported OSS-to-SRAN migration for **3,500+ sites** across 2G/3G/4G, ensuring configuration integrity and minimal service disruption.
- Contributed to nationwide 4G upgrade across **15 states**; provided second-level BSC-level fault troubleshooting and performance analysis.

SanRidge Pvt. Ltd. — Android Development Intern

Hyderabad, India · Dec 2020 – May 2021

- Developed Android applications (Kotlin, Android Studio) with Material Design UI components through full SDLC: design, development, testing, debugging.

TECHNICAL SKILLS

Security	NIST CSF (deep) · MITRE ATT&CK · Threat Modeling · Vulnerability Assessment · Penetration Testing · Intrusion Detection (4 modalities) · Anomaly Detection/ML (IsolationForest, RandomForest) · Attack Surface Management · Dark Web OSINT · Blockchain Forensics · SIEM · Incident Response
Data Healthcare	& HL7 v2 · FHIR R4 · ETL Pipeline Design · SQL · Python (pandas, scikit-learn, scapy) · Tableau · Power BI · Healthcare Interoperability · Data Validation
Engineering Infra	& Python (deep) · FastAPI (deep) · PowerShell · Bash · Linux · Git · Docker · React · REST APIs · Ollama · AWS · Azure · TCP/IP, DNS, DHCP
Telecom / RAN	RAN 2G/3G/4G (deep — 3,500+ site migration, 15 states) · OSS Scripting · XML automation
Tools	Wireshark · Nmap · Nuclei · subfinder · httpx · scapy · Gitleaks · Gemini / OpenAI / Grok / NVIDIA APIs · Tor proxy · Blockchain.info / Etherscan APIs
Languages	Python · PowerShell · SQL · Bash · Kotlin (legacy) · XML

CERTIFICATIONS

Certification	Credential ID	Validity / Date	Issuer
CompTIA Security+ (SY0-701)	COMP001022881385	Oct 2025 – Oct 2028	CompTIA
Fortinet NSE 1 — Network Security Associate	NudBeGrVP0	Jul 13, 2021	Fortinet
Fortinet NSE 2 — Network Security Associate	MBGdIukeds	Jul 20, 2021	Fortinet
Fortinet NSE 4 — FortiGate Security 7.0	Course completion	Jul 21, 2021 · 9.1 CPE hrs	Fortinet
Automation Anywhere — Essentials RPA Professional	AADV6-63718193	On certificate	Automation Anywhere
EV Design & Development	ISIE/2019/SDP/06/KLUCC/EV/01	Feb 26 – Apr 4, 2019	ISIEINDIA / MoNRE GOI
HackerRank Python (Basic)	E3EF8F866323	May 15, 2025	HackerRank
HackerRank Problem Solving (Basic)	A99E37EE6E6C	Sep 15, 2020	HackerRank
HackerRank Problem Solving (Intermediate)	F137B6531CCC	Sep 12, 2020	HackerRank

HONORS & SERVICE

- **Management Conference 2022 — Vodafone Idea Ltd.** (Apr 2022 – Apr 2023) — Organizing committee member; supported operational logistics and stakeholder coordination for company-wide leadership development events.